

AVIRAL NIGAM

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RESEARCH INTERESTS

My focus is AI safety. I study reasoning faithfulness and confidence calibration, and I'm interested in red teaming to uncover model failures. I'm also interested in how reinforcement learning from human feedback (RLHF) shapes behavior, and in instrumental convergence: why AI systems with different goals may develop similar incentives to seek power or preserve themselves.

EDUCATION

Manipal Institute of Technology, MAHE

Manipal, India

B.Tech., Data Science and Engineering

Expected July 2028

- Relevant coursework: Deep Learning, Statistical Modeling, Linear Algebra & Probability, Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems.
- Honors: Shortlisted, Algoverse AI Research Fellowship (Fall 2026 cohort).

PUBLICATIONS

Last Translation Benchmark (LTB) [arXiv:2609.04173](https://arxiv.org/abs/2609.04173)

Sept 2026

V. Zouhar et al. Last Translation Benchmark. *arXiv:2609.04173 [cs.CL]*, 2026. Named co-author on the dataset-author list, ordered by contribution.

- Large-scale, multi-institution benchmark (ETH Zürich, JHU, MBZUI, KIT, Amsterdam, Queen's Belfast, and others): 3,456 human-authored, peer-reviewed examples across 109 languages spanning text, image, audio, and video, built by 244 contributors from 166 institutions, each paired with handcrafted verification rules for objective scoring where automatic metrics saturate.
- Designed the adversarial multimodal examples used to construct the benchmark, directly contributing to results showing that even the best current model (GPT-6 Astra Pro) passes just 45.33% of examples on the live leaderboard, against 99.89% for human translators - evidence that headline benchmark scores do not survive targeted adversarial construction.

Dataset: huggingface.co/datasets/zouhar/last-translation-benchmark | Leaderboard: last-translation-benchmark.vilda.net

AI SAFETY & EVALUATION RESEARCH

FactTrace - Chain-of-Thought Faithfulness Monitor github.com/Aviral0gHub/facttrace

Aug 2026

- Claim-level factuality verifier for agentic LLM reasoning traces: decomposes multi-step reasoning into atomic claims, retrieves evidence via FAISS dense retrieval, and labels each claim supported, contradicted, or unverifiable using NLI entailment scoring - black-box, requiring no model internals.
- Improved claim-level accuracy from 78.5% (lexical baseline) to 87.7% on a 65-claim curated benchmark by replacing regex heuristics with dense retrieval and RoBERTa-large-MNLI - a 9-point swing showing how much of a faithfulness result is an artifact of metric design.
- Found unsupported claims concentrate in later reasoning steps, precisely where a trace-based monitor is least likely to catch them; benchmark size and single-model scope documented as stated limitations.

CalibCheck - Calibration & Honesty Evaluation github.com/Aviral0gHub/calibcheck

Aug 2026

- Epistemic-evaluation pipeline measuring whether a model's stated confidence tracks its accuracy: verbalized self-reported confidence vs. raw next-token probability, scored by Expected Calibration Error, Brier score, and reliability diagrams across 6 topic domains.
- Found Qwen2.5-0.5B-Instruct overconfident by 14.3 points on average (ECE = 0.14) over 60 questions, with domain-conditional miscalibration up to 57.5 points on sports (30% accuracy vs. 87.5% stated confidence); small-n variance caveats reported explicitly.

TECHNICAL SKILLS

Evaluation & AI Safety Behavioral and black-box evals, calibration (ECE, Brier, reliability diagrams), claim decomposition, retrieval-augmented verification, NLI/entailment scoring, adversarial benchmark construction

ML & Models PyTorch, HuggingFace Transformers, Sentence-Transformers, scikit-learn, FAISS, metric & transfer learning; Qwen2.5-0.5B-Instruct, RoBERTa-large-MNLI, ResNet

Languages & Tooling Python, Java, Shell/Bash, Linux, Git, CUDA